

```
#Importing Necessary Libraries for EDA
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

#importing data into the environment
data = pd.read_csv('data.csv')

#Basic Info
# Show top rows
print(data.head())

# Overview of the dataset
print(data.info())

# Summary statistics (including categorical data)
print(data.describe(include='all'))

# Check for missing values
print(data.isnull().sum())

#drop unnecessary columns ('availableCountries' in this case as 99% of the rows there are missing)
data = data.drop(columns=['availableCountries'])
```

```

0          Ariel  movie  Comedy, Crime, Romance  1988.0 \
1  Shadows in Paradise  movie  Comedy, Drama, Music  1986.0
2      Forrest Gump  movie  Drama, Romance  1994.0
3    American Beauty  movie  Drama  1999.0
4    The Fifth Element  movie  Action, Adventure, Sci-Fi  1997.0

      imdbId  imdbAverageRating  imdbNumVotes  availableCountries
0  tt0094675              7.4         9121.0                NaN
1  tt0092149              7.4         7934.0                NaN
2  tt0109830              8.8        2369309.0                NaN
3  tt0169547              8.3        1246137.0                NaN
4  tt0119116              7.6         525576.0                NaN
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 21398 entries, 0 to 21397
Data columns (total 8 columns):
#   Column                Non-Null Count  Dtype
---  -
0   title                 20682 non-null  object
1   type                 21398 non-null  object
2   genres               21026 non-null  object
3   releaseYear          21356 non-null  float64
4   imdbId               19788 non-null  object
5   imdbAverageRating    19597 non-null  float64
6   imdbNumVotes         19597 non-null  float64
7   availableCountries    161 non-null   object
dtypes: float64(3), object(5)
memory usage: 1.3+ MB
None

      title  type  genres  releaseYear  imdbId  imdbAverageRating \
count    20682  21398   21026  21356.000000    19788    19597.000000
unique    19857      2     869         NaN    19784         NaN
top      Animal  movie  Comedy         NaN  tt31805297         NaN
freq         5  16158   1790         NaN         2         NaN
mean         NaN     NaN     NaN    2013.320519         NaN     6.400939
std         NaN     NaN     NaN     14.300791         NaN     1.092405
min         NaN     NaN     NaN    1913.000000         NaN     1.200000
25%         NaN     NaN     NaN    2012.000000         NaN     5.800000
50%         NaN     NaN     NaN    2018.000000         NaN     6.500000
75%         NaN     NaN     NaN    2022.000000         NaN     7.200000
max         NaN     NaN     NaN    2027.000000         NaN     9.800000

      imdbNumVotes  availableCountries
count  1.959700e+04                161
unique         NaN                 99
top         NaN  AD, AE, AG, AL, AO, AR, AT, AU, AZ, BA, BB, BE...
freq         NaN                 11
mean  3.174670e+04                NaN
std   1.206873e+05                NaN
min   5.000000e+00                NaN
25%   3.290000e+02                NaN
50%   1.599000e+03                NaN
75%   1.018400e+04                NaN
max   3.031990e+06                NaN
title                                716
```

```
type          0
genres        372
releaseYear   42
```

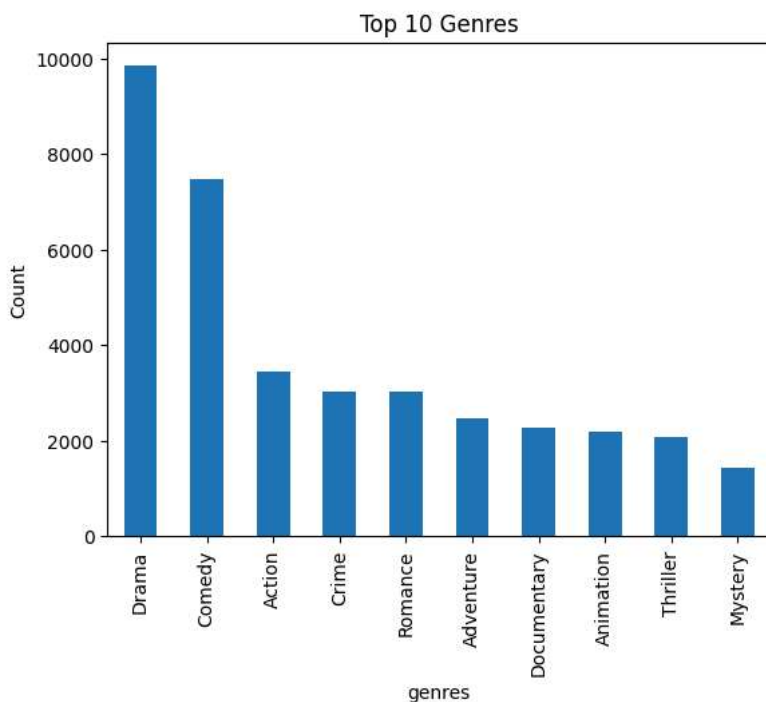
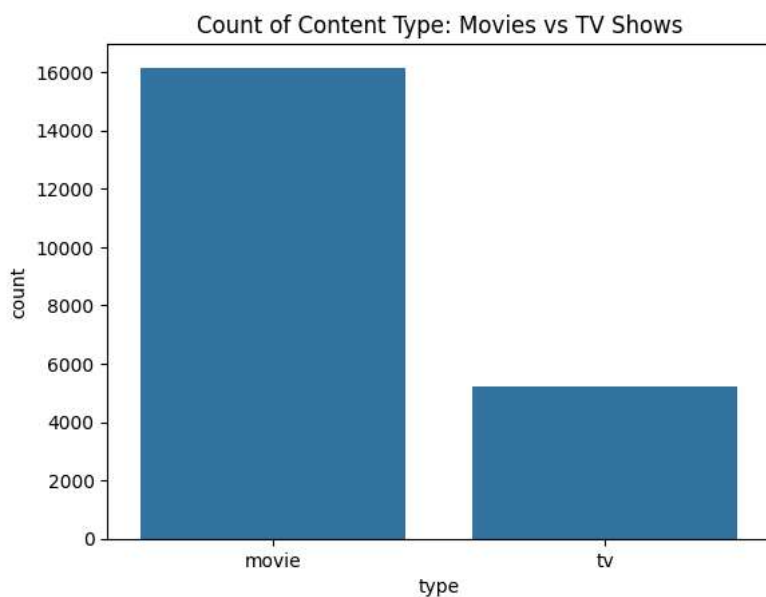
```
#Univariate Analysis
```

```
#Showing the count of movies vs tv shows
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```
sns.countplot(x='type', data=data)
plt.title('Count of Content Type: Movies vs TV Shows')
plt.show()
```

```
#Top 10 genres
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```
data['genres'].str.split(', ').explode().value_counts().head(10).plot(kind='bar')
plt.title('Top 10 Genres')
plt.ylabel('Count')
plt.show()
```



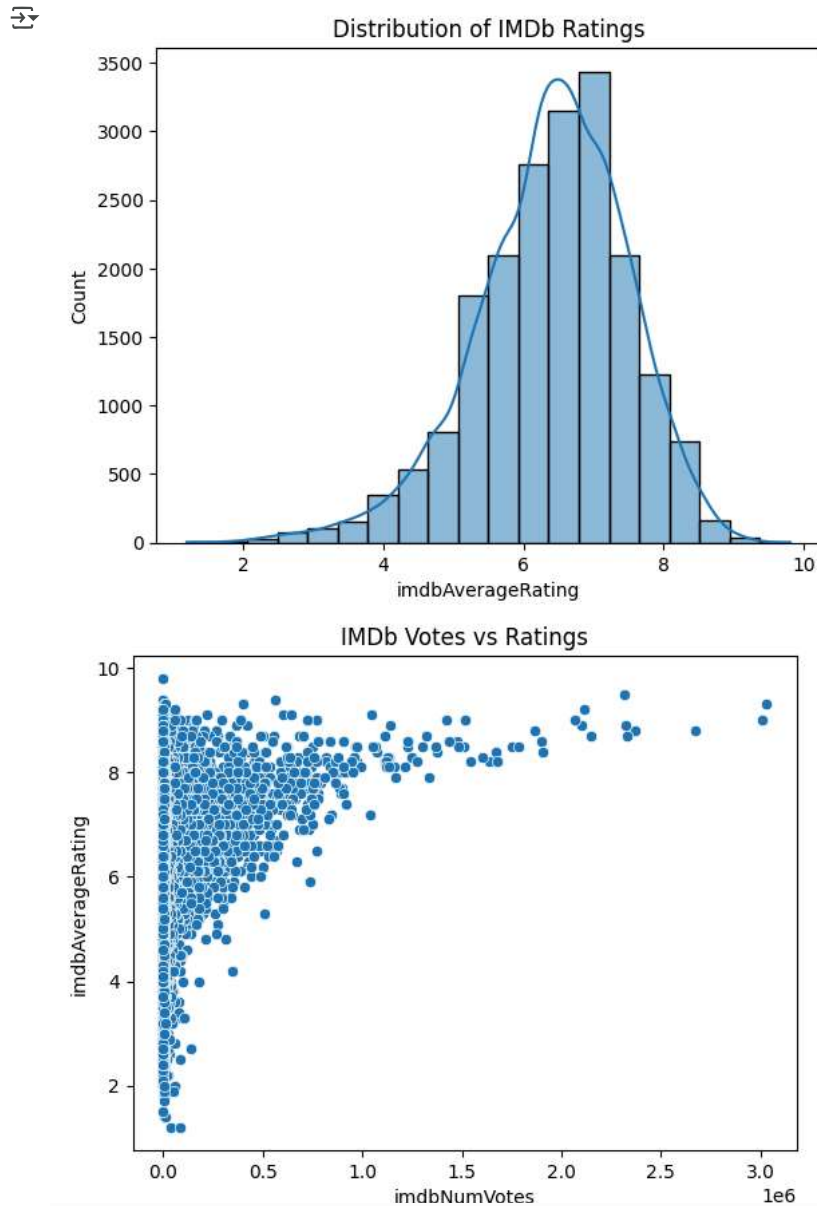
```
# Distribution of IMDb Ratings
```

```
sns.histplot(data['imdbAverageRating'], bins=20, kde=True)
plt.title('Distribution of IMDb Ratings')
plt.show()
```

```
# Votes vs Ratings
```

```
sns.scatterplot(x='imdbNumVotes', y='imdbAverageRating', data=data)
```

```
plt.title('IMDb Votes vs Ratings')
plt.show()
```

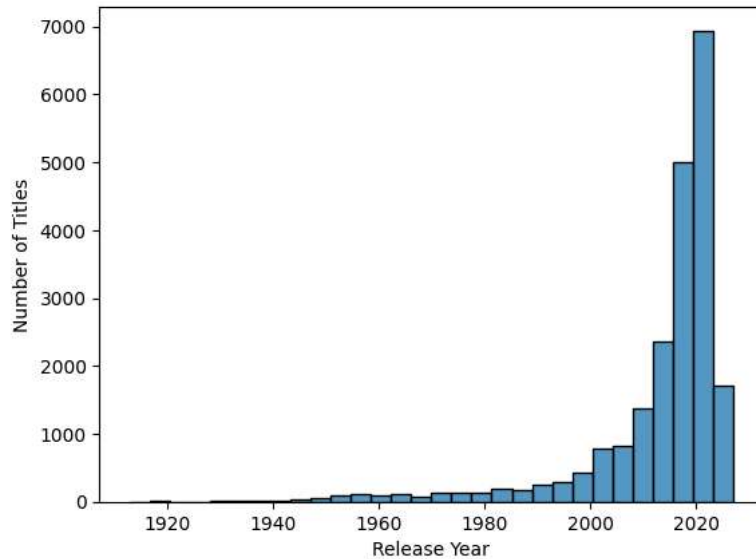


```
# Content count by release year
sns.histplot(data['releaseYear'].dropna(), bins=30, kde=False)
plt.title('Content Released Over the Years')
plt.xlabel('Release Year')
plt.ylabel('Number of Titles')
plt.show()
```

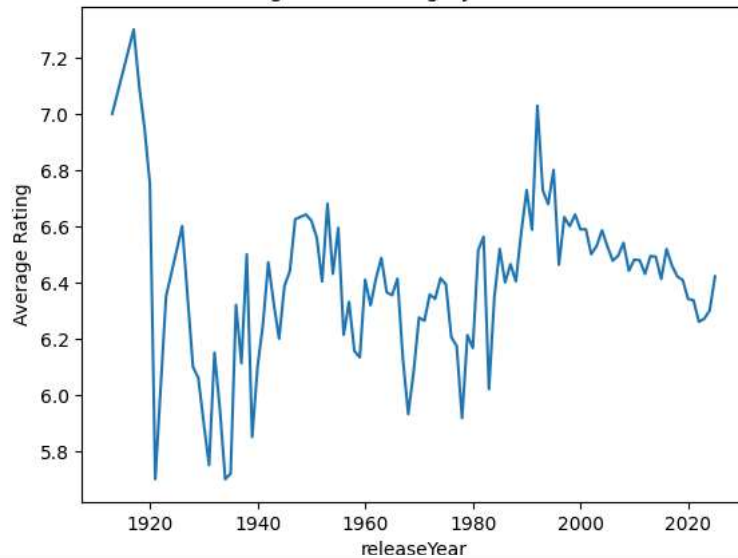
```
# Average rating by year
data.groupby('releaseYear')['imdbAverageRating'].mean().plot()
plt.title('Average IMDb Rating by Release Year')
plt.ylabel('Average Rating')
plt.show()
```



Content Released Over the Years



Average IMDb Rating by Release Year



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EDA Summary - Netflix Dataset

1. Content Type Distribution

The dataset contains two types of content: Movies and TV Shows.

Movies clearly dominate the platform, appearing significantly more than TV shows.

2. Top Genres

After exploding multi-genre entries, the most common genres are:

Drama

Comedy

Action

Romance

Thriller

These genres reflect Netflix's focus on engaging, emotionally-driven content

with mass appeal.

3. IMDb Ratings Distribution

Ratings are mostly clustered between 6.0 and 8.5, with a slight right-skew.

A few titles have exceptionally high ratings (8.5+), but these are rare.

Very low-rated content (below 5.0) is minimal, indicating overall decent quality.

4. Votes vs Ratings

There's a positive but loose correlation between IMDb votes and ratings.

Highly rated titles often attract more votes, but there are also well-rated titles with fewer votes (niche or new releases).

Outliers exist: some titles have high votes but only moderate ratings, possibly due to hype or controversy.

5. Content Released Over the Years

The number of content releases gradually increases, especially from the 2000s onward.

A noticeable spike in content appears post-2010, showing Netflix's aggressive content expansion era.

6. Average IMDb Rating by Release Year

Average IMDb ratings have remained fairly stable over the decades.

Slight fluctuations exist year to year, but no clear long-term upward or downward trend is observed.

This implies that while quantity has increased, Netflix has managed to maintain consistent quality.

Overall Insights:

Netflix heavily favors movies over TV shows.

Popular genres like Drama, Comedy, and Action dominate the platform.

Audience ratings are generally favorable, and the platform's content production has surged in recent years.

There's no clear trend that newer content is better or worse rated – Netflix maintains a consistent quality bar.

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